# Shot Transition Detection Using a Perceptual Distance Function

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## Abstract

A shot transition occurs when a frame fails to maintain perceptual continuity with its previous frame. To maintain perceptual continuity, two consecutive frames must contain similar perceptual features. In this paper, we propose using the dynamic partial function (DPF) as the inter-frame distance function to detect perceptual discontinuity, and hence the boundary of a shot. Through analysis and experiments, we show that DPF is more effective than traditional distance functions for detecting shot transitions. We also discuss the foundation of DPF in cognitive psychology.

## 1 Introduction

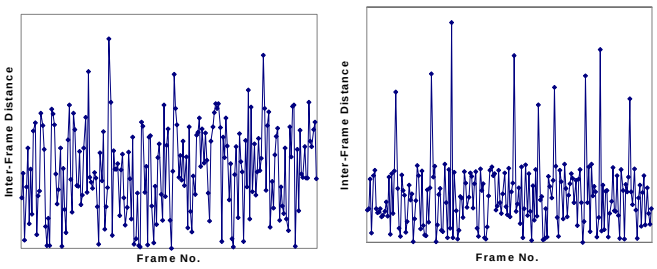
Video *shot-transition* detection is a critical research topic, since reliably recovering a *shot*, the elementary video unit, provides the foundation for nearly all high-level video segmentation/analysis algorithms [8]. A large body of work has been conducted on this research topic, but there is still room for improvement.

A typical shot-transition detecting technique involves in two stages:

1. *Extracting intra-frame features.* Intra-frame features are features of individual video frames (i.e., still images). As depicted in [11], widely used image features include colors, intensities, textures, shapes, and layout information.

2. *Comparing inter-frame distance.* Videos differ from still images in that they carry both spatial and temporal changes. A shot transition takes place at where inter-frame distance is *perceptually significant.*

Given intra-frame features, the heart of the shot-transition detection research is to devise a perceptual distance function which can reliably capture the perceptual discontinuity [4, 5] at shot boundaries. We use Figure 1 to illustrate what a good distance function should accomplish versus what a poor distance function might do. The x-axis of the figure depicts the frame number; the $y$-axis the distance between the $i^{th}$ and the $i+1^{th}$ frames. Figure 1(a) shows a poor distance function, which fails to separate the perceptually dissimilar frames (shot boundaries) from the perceptually similar ones. Figure 1(b) shows a good distance function, which successfully separates the large distances at the shot boundaries from the small distances at the other frame boundaries.




(a) A Poor Function (b) A Good Function

Figure 1: The Effects of Distance Functions.

In this paper, we propose using the dynamic partial function [7] (DPF) for detecting shot boundaries. As its name suggests, DPF *dynamically* selects a *partial* feature subset to compare frames. DPF works effectively for the following two reasons:

1. *Partial.* Two frames that belong to the same shot should have a significant number of similar intra-frame features for maintaining perceptual continuity. Object movement and camera motions affect only a subset of the features. Once these features are discounted by DPF, the remaining features should be perceptually similar, or a shot transition will take place between the frames.

2. *Dynamic.* DPF always discounts the largest fea-

ture distances, and thus it can dynamically ignore the influence of different features caused by different spatial or temporal changes. For instance, a *pan* might alter only color features. A *zoom* might affect only texture channels. DPF dynamically removes the largest feature distances, and hence also those features that do not help maintain perceptual continuity.

In summary, human perception maintains continuity when we perceive a sufficient number of similar features from one frame to the next. DPF removes the largest dissimilar features. If the remaining features are not similar, perception continuity cannot be maintained, and therefore a shot transition will take place. Our empirical study shows that DPF achieves superior shot-detection accuracy over traditional distance functions.

The rest of this paper is organized into three sections. Section 2 presents DPF. Section 3 summarizes the experimental results. We offer concluding remarks in Section 4.

## 2 Dynamic Partial Function

Given a set of intra-frame features, our goal is to design a distance function that can accurately determine shot boundaries.

The Minkowski metric is widely used in shot boundary detection algorithms. Suppose two objects $X$ and $Y$ are represented by two $p$ dimensional vectors $(x_1, x_2, \cdots, x_p)$ and $(y_1, y_2, \cdots, y_p)$, respectively. The Minkowski metric $d(X, Y)$ is defined as

$$d(X,Y) = (\sum_{i=1}^{p} |x_i - y_i|^r)^{\frac{1}{r}}, \qquad (1)$$

where $r$ is the Minkowski factor for the norm. Particularly, when $r$ is set as 2, it is the well-known Euclidean distance; when $r$ is 1, it is the Manhattan distance (or $L_1$ distance). Objects that are similar will be closer together in the feature space than objects that mismatch on a number of features [10].

Measuring similarity by the Minkowski metric is based on one assumption: that two frames are similar when the aggregated distance between all their features is similar. However, this assumption runs counter to the way human eyes maintain perceptual continuity. The perceptual continuity is maintained because *constancy* is maintained in a large fraction of the feature-channels [4, 5]. For instance, feature vector $V_1 = (0.5, 0.5, 0.1)$ and feature vector $V_2 = (0.5, 0.5, 0.9)$ can have good perceptual continuity since the majority of the features are stable. On the other hand, feature vector $V_3 = (0.3, 0.3, 0.2)$ and feature vector $V_4 = (0.5, 0.5, 0.9)$ are perceptually discontinuous, since all features are different. (Frame $V_4$ has no stable reference on frame $V_3$.) However, the Euclidean distance function would characterize that the $V_3$ - $V_4$ pair as being more similar to the $V_1$ - $V_2$ pair.

Based on the observations explained above, we designed a distance function to better characterize perceptual continuity. We define the *feature distance* on the $i^{th}$ feature as

$$\delta_i = |x_i - y_i|. \qquad (i = 1, \cdots, p)$$

We define sets $\Delta_m$ as

$$\Delta_m = \{The \;\; smallest \;\; m \; \delta's \;\; of \;\; (\delta_1, ..., \delta_p)\}.$$

DPF can then be expressed as

$$d(m,r) = (\sum_{\delta_i \in \Delta_m} {\delta_i}^r)^{\frac{1}{r}}. \qquad (2)$$

DPF has two adjustable parameters: $m$ and $r$. Parameter $m$ can range from 1 to $p$. When $m = p$, it degenerates to the Minkowski metric. When $m < p$, it counts only the smallest $m$ feature distances between two frames, and the influence of the $(p - m)$ largest feature distances is eliminated. Note that DPF dynamically selects features to be considered for different pairs of frames. This is achieved by the introduction of $\Delta_m$, which changes dynamically for different pairs of frames. In Section 3, we will show that if a proper value of $m$ is chosen, we can better single out frames that cause perceptual discontinuity.

## 3 Experiments

Our testbed consists of 150 video clips downloaded from the Internet. The videos cover the following subjects:

- Cartoon: 30 clips, each clip lasting for 50 seconds (from commercial CDs).
- Documentary: 70 clips, each lasting for two to five minutes [2].
- Comedy: 50 clips, each lasting for up to 30 seconds [1].

For characterizing a frame, we extract a set of 144 intra-frame features. These features include color histograms, color means, color variances, color spreadness, color-blob elongation, and texture features in three orientations (vertical, horizontal, and diagonal) and three resolutions (coarse, medium, and fine) [6, 12].

Our experiments have two goals. The first one is to find the optimal parameter $m$ settings for DPF (Section 3.1). The second goal is to compare the shot detection accuracy between employing DPF and employing the Minkowski metric as the inter-frame distance function (Section 3.2).

### 3.1 Parameter $m$

DPF has two parameters, $r$ and $m$. We fix $r = 2$ in our empirical study, since we compare only with Euclidean distance function (perhaps the most widely used distance function), which sets $r = 2$. To find a good $m$, we use a machine learning approach. We sample 40% of the video clips as the training data to discover a good $m$. We then use the remaining 60% of video clips as testing data to examine the effectiveness of the learned $m$.

In the training phase, we label the accurate positions of shot boundaries. We then experiment with different values of $m$ on three video datasets (cartoon, documentary, and comedy). Figure 2 shows that for all three video types, the false detection rates are reduced to a minimum as $m$ is reduced from 144 to between 115 and 120. Note that when $m = 144$, DPF is degenerated into the Euclidean function. It is evident that the Euclidean distance function is not the best choice for our purpose.

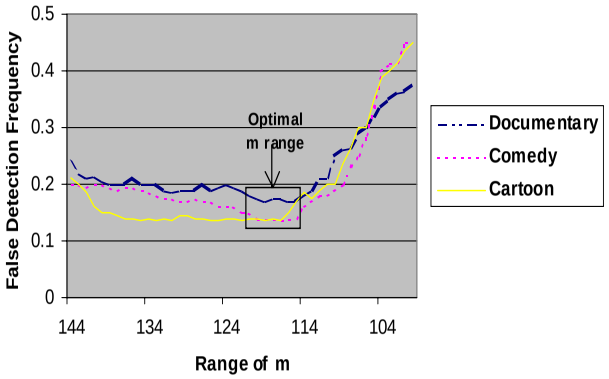



Figure 2: Optimal $m$.

### 3.2 DPF vs. Minkowski

We next compare two inter-frame functions: DPF and Euclidean. For DPF, we set $m = 117$. Table 1 shows that DPF improves the detection accuracy over the Euclidean distance function on both precision and recall for all video categories. The average improvement as shown in Figure 3 is about 7% in both recall and precision.

Figure 4 illustrates why DPF can better detect shot boundaries than Euclidean distance from the signal/noise ratio perspective. The x-axis of the figure depicts the frame number; the y-axis the inter-frame distance between the $i^{th}$ and the $i + 1^{th}$ frames. We mark each real shot boundary with a circle and a false detection with a cross. Figure 4(a) shows that the Euclidean distance function identifies four shot boundaries, in which the left-most one is a false positive. Figure 4(b) shows that DPF separates the distances between shot boundaries and non-boundaries better, and hence eliminates the one mis-detection. DPF improves the signal/noise ratio, and therefore, it is more effective in detecting shot transitions.

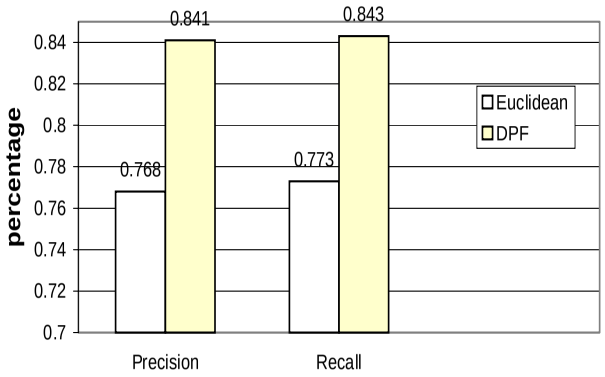
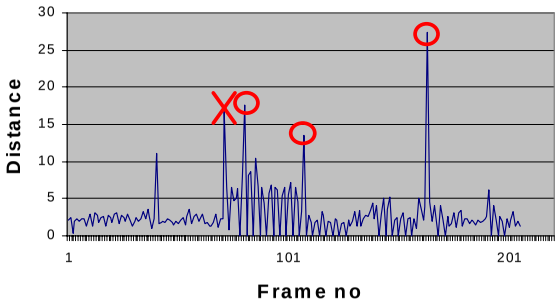
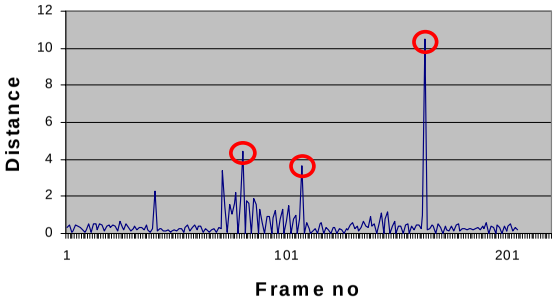



Figure 3: Overall Error Comparison.




(a) Euclidean




(b) DPF

Figure 4: Euclidean vs. DPF.

| *Distance Functions* | *Video Type* | Comedy | *Cartoon* | *Documentary* |
|---|---|---|---|---|
| | *# of Shot Boundaries* | 425 | 167 | 793 |
| *Euclidean* | *# of false* | 93 | 39 | 192 |
| | *# of miss* | 97 | 37 | 183 |
| | *Precision* | 78.1% | 76.6% | 75.8% |
| | *Recall* | 77.2% | 77.8% | 76.9% |
| *DPF* | *# of false* | 61 | 26 | 140 |
| | *# of miss* | 67 | 25 | 129 |
| | *Precision* | 85.6% | 84.4% | 82.3% |
| | *Recall* | 84.2% | 85.0% | 83.7% |

Table 1: Precision and Recall.

## 4 Conclusion

Most of the previous shot-detection algorithms use a Minkowski-like metric as the inter-frame distance function. In this paper, we have explained that the Minkowski-like metric may not be able to capture perceptual discontinuity most effectively for shot-transition detection. We have introduced DPF, which discounts the changes (likely to be spatial and temporal changes) between frames. If the remaining features are still quite different, then probably the two frames do not pass perceptual continuity and hence DPF declares a shot boundary. Our empirical study has shown that DPF outperforms the Minkowski metric in both precision and recall on different types of video clips.

The result of our empirical study is encouraging partly because DPF demonstrates its effectiveness, and partly because our result confirms the following theories of cognition:

**1**. The axioms of metric spaces (e.g., minimality, symmetry, and the triangle inequality) appear to be violated by human similarity judgements [9, 13].

**2**. Two objects (i.e., two frames) are similar (and hence perceptual continuity is maintained) because they share some common features (the partial aspect of DPF), and the features for determining similarity between two objects may be developed or introduced during similarity judgements (the dynamic aspect of DPF) [3].

The focus of this paper has been the study of inter-frame distance functions. We have not evaluated more sophisticated methods (such as the double-thresholding method) to identify shot boundaries of different types (hard cut, fade, wipe, and dissolve). In our future work, we will evaluate the effectiveness of DPF on transition-type identification.

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